

Database Design Document

Online Shopping System - E-Commerce Platform

1. Overview

This document describes the database schema for an e-commerce platform. The system uses MySQL 8.0 with InnoDB engine for ACID compliance. The schema supports user management, product catalog, order processing, and payment tracking.

2. Entity Relationship

Core entities: Users, Products, Categories, Orders, OrderItems, Payments, Reviews, Addresses. Relationships follow 3NF normalization.

3. Table Definitions

3.1 Users Table

Column	Type	Constraints	Description
id	BIGINT	PK, AUTO_INCREMENT	User ID
username	VARCHAR(50)	UNIQUE, NOT NULL	Login name
email	VARCHAR(100)	UNIQUE, NOT NULL	Email address
password_hash	VARCHAR(255)	NOT NULL	Bcrypt hash
phone	VARCHAR(20)	NULL	Phone number
status	ENUM	DEFAULT 'active'	active/disabled/banned
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Registration time

3.2 Products Table

Column	Type	Constraints	Description
id	BIGINT	PK, AUTO_INCREMENT	Product ID
name	VARCHAR(200)	NOT NULL	Product name
description	TEXT	NULL	Full description
price	DECIMAL(10,2)	NOT NULL, CHECK > 0	Unit price
stock	INT	DEFAULT 0, CHECK >= 0	Available stock
category_id	BIGINT	FK -> categories.id	Category
is_active	BOOLEAN	DEFAULT TRUE	Listing status

4. Index Strategy

- B-Tree index on users.email for login queries
- Composite index on orders(user_id, status) for order listing
- Full-text index on products.name + products.description for search
- Covering index on order_items(order_id, product_id, quantity, price)

5. SQL Scripts

```
CREATE TABLE users ( id BIGINT AUTO_INCREMENT PRIMARY KEY, username  
VARCHAR(50) NOT NULL UNIQUE, email VARCHAR(100) NOT NULL UNIQUE,  
password_hash VARCHAR(255) NOT NULL, created_at TIMESTAMP DEFAULT  
CURRENT_TIMESTAMP ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```